

Total No. of Questions : 6]

SEAT No. :

P47

[Total No. of Pages : 2

Oct.-16/T.E./Insem.-47
T.E. (Information Technology)
DATABASE MANAGEMENT SYSTEM
(2012 Pattern) (Semester - I)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6.*
- 2) Neat diagrams must be drawn wherever necessary.*
- 3) Figures to the right indicate full marks.*
- 4) Assume suitable data, if necessary.*

- Q1)** a) Define the terms entity attribute, role & relationship between the entities, giving examples for each of them. [5]
- b) Define Armstrong's axioms for functional dependency. [5]

OR

- Q2)** a) Define dependency preserving & loss less join decomposition. [4]
- b) Write a short note on : [6]
- Mapping of IS-A relationship of ER diagram to tables.
- Q3)** a) Discuss with respect to SQL : [6]
- i) How nulls are treated in comparison operator?
 - ii) The check clause/constraint
 - iii) The difference between drop table R & delete from R.
- b) Explain in brief the advantages of mongoDB over RDBMS. [4]

OR

- Q4)** a) Why are the cursors necessary in embedded SQL? [2]

P.T.O.

- b) Consider the insurance database, where the primary keys are underlined: [8]
person (SS#, name, address)
car (license, year, model)
accident (date, driver, damage-amount)
owns (SS#, License)
log (license, date, driver)
Construct the SQL queries :
i) Find the total no. of people whose cars were involved in accidents in 2010.
ii) Find the no. of accidents in which the cars belonging to 'John Smith' were involved.
iii) Add a new customer to the database.
iv) Add a new accident record for Toyota belonging to 'Jones'.

- Q5)** a) What is a checkpoint? List the operations to be performed by the system when a checkpoint is to be taken. What does the recovery system do if there is a crash? [6]
b) When are two schedules said to be view equivalent? [4]

OR

- Q6)** a) Define granularity hierarchy of granularity of locks & multiple granularity locking. Describe the modified two phase locking with multiple granularity locking. [6]
b) What are the costs to be considered when a transaction has to be rolled back when recovering from deadlock? [4]

